

# AP CALCULUS AB · THE UNIVERSAL STUDY GUIDE

Targeting a 3, 4, or 5 — every concept tagged so you know what your tier needs

Tier badges: [3] foundational · [4] adds for 4+ · [5] adds for 5

Targeting a 4 → study [3] AND [4] items. Targeting a 5 → study all three tiers.

## PAGE 1 | MATH OF 3/4/5 · UNITS · MASTER THEOREMS

### THE MATH OF YOUR TARGET

| Target      | MCQ (45 q, 50%) | FRQ (6 q, 50%)  | Approx % | Tier focus |
|-------------|-----------------|-----------------|----------|------------|
| 3 (PASS)    | ~22 / 45 (~49%) | Earn ~25/54 raw | ~45-50%  | [3] only   |
| 4 (SOLID)   | ~28 / 45 (~62%) | Earn ~33/54 raw | ~60-65%  | [3] + [4]  |
| 5 (MASTERY) | ~34 / 45 (~76%) | Earn ~41/54 raw | ~73-78%  | All three  |

### THE 8 UNITS — by exam weight

| # | Unit                                          | Weight | Key topics                                                  |
|---|-----------------------------------------------|--------|-------------------------------------------------------------|
| 1 | Limits & Continuity                           | 10–12% | Limit laws, asymptotes, continuity, IVT                     |
| 2 | Differentiation: Definition + Basic Rules     | 10–12% | Limit definition, power/product/quotient rule               |
| 3 | Differentiation: Composite, Implicit, Inverse | 9–13%  | Chain rule, implicit, inverse trig, log diff                |
| 4 | Contextual Applications of Differentiation    | 10–15% | <b>Related rates</b> , motion, linearization                |
| 5 | Analytical Applications of Differentiation    | 15–18% | <b>Optimization</b> , MVT, EVT, concavity, curve sketching  |
| 6 | Integration & Accumulation of Change          | 17–20% | <b>FTC</b> , antiderivatives, u-substitution, Riemann sums  |
| 7 | Differential Equations                        | 6–12%  | Separable equations, slope fields, exponential growth/decay |
| 8 | Applications of Integration                   | 10–15% | Area between curves, volumes (disks/washers), average value |

Units 5 and 6 are 32–38% combined. Optimization, MVT, and integration techniques dominate the FRQs.

### THE 5 MASTER THEOREMS — name them in your FRQ justifications [4]/[5]

IVT Intermediate Value Theorem [3]

If  $f$  is CONTINUOUS on  $[a,b]$  and  $N$  is between  $f(a)$  and  $f(b)$ , then there exists  $c$  in  $(a,b)$  with  $f(c) = N$ .

USE: prove a value/zero exists.

EVT Extreme Value Theorem [4]

If  $f$  is CONTINUOUS on a CLOSED interval  $[a,b]$ , then  $f$  attains its absolute max and min on  $[a,b]$ .

USE: justify existence of extrema on closed intervals.

MVT Mean Value Theorem [4]

If  $f$  is CONTINUOUS on  $[a,b]$  and DIFFERENTIABLE on  $(a,b)$ , then there exists  $c$  in  $(a,b)$  with  $f'(c) = (f(b) - f(a))/(b - a)$ .

USE: prove average rate of change is achieved instantaneously.

FTC PART 1 Fundamental Theorem of Calculus [3]

$\frac{d}{dx} \left[ \int_a^x f(t) dt \right] = f(x)$

USE: derivative of an integral with variable upper limit

FTC PART 2 Evaluation [3]

$\int_a^b f'(x) dx = f(b) - f(a)$

USE: compute definite integrals; "net change" interpretation

**KEY INSIGHT** [4]/[5] When using ANY theorem in an FRQ justification, name it explicitly. "By the Mean Value Theorem..." or "By the IVT, since  $f$  is continuous on  $[1,4]$ ..." Graders look for the named theorem; missing the name often costs the justification point.

## UNIT 1 | LIMITS &amp; CONTINUITY

## Limit Laws — the foundation [3]

If  $\lim f(x) = L$  and  $\lim g(x) = M$ :

$$\begin{aligned}\lim [f \pm g] &= L \pm M \\ \lim [f \cdot g] &= L \cdot M \\ \lim [f / g] &= L / M \quad (\text{provided } M \neq 0) \\ \lim [c \cdot f] &= c \cdot L \\ \lim [f(x)]^n &= L^n\end{aligned}$$

## Evaluating Limits — strategy [3]/[4]

- **[3] DIRECT SUBSTITUTION** — try this first. If  $f(c)$  is defined, the limit usually equals  $f(c)$  (for continuous functions).
- **[3] FACTOR + CANCEL** — for  $0/0$  indeterminate forms. Factor numerator and denominator, cancel common factor, then substitute. Example:  
 $\lim_{x \rightarrow 2} (x^2 - 4)/(x - 2) = \lim (x + 2) = 4$ .
- **[4] CONJUGATE TRICK** — for limits with square roots in numerator or denominator. Multiply by conjugate over itself.
- **[4] L'HÔPITAL'S RULE** — for  $0/0$  or  $\infty/\infty$  forms:  $\lim f/g = \lim f'/g'$ . Apply repeatedly if needed.
- **[5] SQUEEZE THEOREM** — if  $g(x) \leq f(x) \leq h(x)$  and  $\lim g = \lim h = L$ , then  $\lim f = L$ . Used for tricky trig limits like  $\lim x \sin(1/x)$ .

## Special Limits to Memorize [3]/[4]

$$\begin{aligned}\lim_{x \rightarrow 0} \sin(x) / x &= 1 & [3] \\ \lim_{x \rightarrow 0} (1 - \cos(x)) / x &= 0 & [4] \\ \lim_{x \rightarrow 0} (1 - \cos(x)) / x^2 &= 1/2 & [5] \\ \lim_{x \rightarrow \infty} (1 + 1/x)^x &= e & [4]\end{aligned}$$

Behavior at infinity (rational functions):

$\deg(\text{num}) < \deg(\text{denom}) \rightarrow \text{limit} = 0$   
 $\deg(\text{num}) = \deg(\text{denom}) \rightarrow \text{limit} = \text{ratio of leading coefficients}$   
 $\deg(\text{num}) > \deg(\text{denom}) \rightarrow \text{limit} = \pm\infty$

## Continuity — 3 conditions [3]

$f$  is CONTINUOUS at  $x = c$  iff:

1.  $f(c)$  is defined
2.  $\lim_{x \rightarrow c} f(x)$  exists
3.  $\lim_{x \rightarrow c} f(x) = f(c)$

TYPES of discontinuity:

- Removable (hole) — limit exists, but  $f(c)$  undefined or  $\neq$  limit
- Jump — left and right limits differ
- Infinite — limit =  $\pm\infty$  (vertical asymptote)

## Asymptotes [4]

- **Vertical asymptote at  $x = c$**   $\Leftrightarrow \lim_{x \rightarrow c} f(x) = \pm\infty$ . Find by: where denominator = 0 AND numerator  $\neq$  0.
- **Horizontal asymptote  $y = L$**   $\Leftrightarrow \lim_{x \rightarrow \pm\infty} f(x) = L$ . Use degree comparison rules above.

## UNITS 2–3 | DIFFERENTIATION · the rules

### The Definition of the Derivative [3]/[4]

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad [\text{DIFFERENCE QUOTIENT}]$$

Equivalent form for derivative AT a point a:

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

GEOMETRIC MEANING: slope of the tangent line to  $f$  at  $x$ .

PHYSICAL MEANING: instantaneous rate of change of  $f$ .

### The Basic Differentiation Rules [3]

POWER RULE:  $\frac{d}{dx} [x^n] = n \cdot x^{n-1}$

CONSTANT RULE:  $\frac{d}{dx} [c] = 0$

CONSTANT MULT:  $\frac{d}{dx} [c \cdot f] = c \cdot f'$

SUM/DIFFERENCE:  $\frac{d}{dx} [f \pm g] = f' \pm g'$

PRODUCT RULE:  $\frac{d}{dx} [f \cdot g] = f' \cdot g + f \cdot g'$

QUOTIENT RULE:  $\frac{d}{dx} [f/g] = (f' \cdot g - f \cdot g') / g^2$   
[memory: "low d-high minus high d-low, square the bottom and away we go"]

CHAIN RULE:  $\frac{d}{dx} [f(g(x))] = f'(g(x)) \cdot g'(x)$   
["derivative of outer at inner, times derivative of inner"]

### Derivatives to Memorize [3]/[4]

TRIG FUNCTIONS:

$$\frac{d}{dx} [\sin x] = \cos x \quad \frac{d}{dx} [\cos x] = -\sin x$$

$$\frac{d}{dx} [\tan x] = \sec^2 x \quad \frac{d}{dx} [\cot x] = -\csc^2 x$$

$$\frac{d}{dx} [\sec x] = \sec x \cdot \tan x \quad \frac{d}{dx} [\csc x] = -\csc x \cdot \cot x$$

EXPONENTIAL & LOG:

$$\frac{d}{dx} [e^x] = e^x \quad \frac{d}{dx} [a^x] = a^x \cdot \ln(a)$$

$$\frac{d}{dx} [\ln x] = 1/x \quad \frac{d}{dx} [\log_a(x)] = 1 / (x \cdot \ln a)$$

INVERSE TRIG:

[4]

$$\frac{d}{dx} [\arcsin x] = 1 / \sqrt{1 - x^2}$$

$$\frac{d}{dx} [\arccos x] = -1 / \sqrt{1 - x^2}$$

$$\frac{d}{dx} [\arctan x] = 1 / (1 + x^2)$$

### Implicit Differentiation [4]

**Use when  $y$  is not isolated.** Take  $d/dx$  of both sides; treat  $y$  as a function of  $x$ ; whenever you differentiate  $y$ , multiply by  $dy/dx$ . Then solve for  $dy/dx$ .

**Worked example:**  $x^2 + y^2 = 25$ . Differentiate:  $2x + 2y \cdot dy/dx = 0 \rightarrow dy/dx = -x/y$ . At point  $(3, 4)$ :  $dy/dx = -3/4$ .

### Logarithmic Differentiation [5]

**Use when:** function is a product/quotient/power of complicated expressions, OR when variable appears in both base AND exponent (like  $x^x$ ).

**Steps:** Take  $\ln$  of both sides  $\rightarrow$  use log properties to expand  $\rightarrow$  differentiate (implicit)  $\rightarrow$  solve for  $dy/dx$ .

**Example:**  $y = x^x$ . Take  $\ln$ :  $\ln y = x \ln x$ . Differentiate:  $(1/y) \cdot dy/dx = \ln x + 1$ . So  $dy/dx = x^x \cdot (\ln x + 1)$ .

**KEY INSIGHT [4]** The chain rule is the most-applied derivative rule on the AP exam. Practice until automatic. Most missed problems involve forgetting to multiply by the inner derivative.

## UNIT 4 | CONTEXTUAL APPLICATIONS · related rates + motion + linearization

### Motion Along a Line — position, velocity, acceleration [3]/[4]

$s(t)$  = position  
 $v(t) = s'(t)$  velocity (signed; sign indicates direction)  
 $a(t) = s''(t) = v'(t)$  acceleration

SPEED =  $|v(t)|$  (always nonneg)  
 DISTANCE TRAVELED on  $[a, b] = \int_a^b |v(t)| dt$   
 DISPLACEMENT on  $[a, b] = \int_a^b v(t) dt = s(b) - s(a)$

PARTICLE SPEEDS UP when  $v$  and  $a$  have SAME SIGN  
 PARTICLE SLOWS DOWN when  $v$  and  $a$  have OPPOSITE SIGNS

### Related Rates — the recipe [4]

**Setup:** quantities change with time; you know one rate, want another. The strategy:

- **Step 1.** Draw a diagram and label all quantities (use VARIABLES, not specific values yet).
- **Step 2.** Write an equation relating the variables (often geometric: Pythagorean theorem, area, volume, similar triangles).
- **Step 3.** Differentiate IMPLICITLY with respect to time  $t$ .
- **Step 4.** ONLY NOW substitute specific values.
- **Step 5.** Solve for the unknown rate. Include UNITS.

### Worked example — ladder sliding [4]

**Problem.** A 13-ft ladder leans against a wall. The bottom slides away from the wall at 2 ft/sec. How fast does the top slide down when bottom is 5 ft from wall?

**Setup.** Let  $x$  = distance from wall,  $y$  = height of top. By Pythagorean:  $x^2 + y^2 = 169$ . When  $x = 5$ ,  $y = 12$ .

**Differentiate w.r.t.  $t$ :**  $2x \cdot dx/dt + 2y \cdot dy/dt = 0 \rightarrow dy/dt = -(x/y) \cdot dx/dt$ .

**Substitute:**  $dy/dt = -(5/12)(2) = -5/6$  ft/sec. **Top slides down at 5/6 ft/sec.**

### Linearization (Tangent Line Approximation) [4]

$$L(x) = f(a) + f'(a) \cdot (x - a) \quad [\text{linearization at } x = a]$$

Use to APPROXIMATE  $f(x)$  for  $x$  near  $a$ .

Example: approximate  $\sqrt{4.1}$  using  $f(x) = \sqrt{x}$  at  $a = 4$ .

$$f(4) = 2, \quad f'(4) = 1/(2\sqrt{4}) = 1/4$$

$$L(4.1) = 2 + (1/4)(0.1) = 2.025 \quad (\text{actual: } \approx 2.0248)$$

### Newton's Method [5]

**Used to approximate a zero of  $f$ .** Iterate:  $x_{n+1} = x_n - f(x_n)/f'(x_n)$ . Start with an initial guess  $x_0$  near the root; converges quickly under good conditions.

**UNIT 5 | ANALYTICAL APPLICATIONS · extrema + concavity + curve sketching****Critical Points + First Derivative Test [3]/[4]**

CRITICAL POINT:  $x = c$  where  $f'(c) = 0$  OR  $f'(c)$  is undefined

FIRST DERIVATIVE TEST (1st DT):

- If  $f'$  changes from POSITIVE to NEGATIVE at  $c \rightarrow$  LOCAL MAX
- If  $f'$  changes from NEGATIVE to POSITIVE at  $c \rightarrow$  LOCAL MIN
- If  $f'$  doesn't change sign  $\rightarrow$  NOT an extremum

The function is INCREASING where  $f' > 0$ , DECREASING where  $f' < 0$

**Concavity + Second Derivative Test [3]/[4]**

$f$  is CONCAVE UP where  $f''(x) > 0$  (cup shape  $\cup$ )  
 $f$  is CONCAVE DOWN where  $f''(x) < 0$  (frown shape  $\cap$ )  
 INFLECTION POINT: where  $f''$  CHANGES SIGN (and  $f$  is continuous)

SECOND DERIVATIVE TEST (2nd DT):

- $f'(c) = 0$  AND  $f''(c) > 0 \rightarrow$  LOCAL MIN
- $f'(c) = 0$  AND  $f''(c) < 0 \rightarrow$  LOCAL MAX
- $f'(c) = 0$  AND  $f''(c) = 0 \rightarrow$  inconclusive (use 1st DT)

**Absolute Extrema on a Closed Interval [4]**

By EVT, on  $[a,b]$  (closed, continuous),  $f$  attains absolute max + min. To find them:

- Find all critical points in  $(a,b)$ .
- Evaluate  $f$  at each critical point AND at the endpoints  $a$  and  $b$ .
- Largest value = absolute max; smallest = absolute min.

**Optimization — the recipe [4]/[5]**

- **Step 1.** Identify what's being optimized (the OBJECTIVE function) and any CONSTRAINT.
- **Step 2.** Use the constraint to express objective as a function of ONE variable.
- **Step 3.** Find critical points by setting derivative = 0.
- **Step 4.** Verify it's a max/min using 1st or 2nd derivative test (or check endpoints).
- **Step 5.** Answer the original question (with units!).

**Worked example — fence problem [4]**

**Problem:** A farmer has 200 ft of fencing. Maximize the area of a rectangular pen against a barn (so only 3 sides need fencing).

**Setup.** Let  $x$  = width perpendicular to barn,  $y$  = length parallel. Constraint:  $2x + y = 200 \rightarrow y = 200 - 2x$ . Objective:  $A(x) = x \cdot y = x(200 - 2x) = 200x - 2x^2$ .

**Differentiate.**  $A'(x) = 200 - 4x$ . Set  $= 0 \rightarrow x = 50$ .

**Verify.**  $A''(x) = -4 < 0$  (always concave down), so  $x = 50$  is a MAX.  $y = 100$ . Maximum area = 5000 ft<sup>2</sup>.

**Mean Value Theorem (MVT) — the existence theorem [4]/[5]**

IF  $f$  is continuous on  $[a,b]$  AND differentiable on  $(a,b)$ ,  
 THEN there exists  $c$  in  $(a,b)$  with  
 $f'(c) = (f(b) - f(a)) / (b - a)$

GEOMETRICALLY: somewhere on  $(a,b)$ , the tangent line is parallel to the secant line.

USE THE NAME ON FRQs: "By the MVT, since  $f$  is continuous on  $[a,b]$  and differentiable on  $(a,b)$ , ..."

## UNIT 6 | INTEGRATION · antiderivatives + FTC + Riemann sums

## Antiderivatives — basic rules [3]

## POWER RULE FOR INTEGRALS:

$$\int x^n dx = x^{(n+1)/(n+1)} + C \quad (\text{for } n \neq -1)$$

$$\int x^{-1} dx = \ln|x| + C$$

## COMMON ANTIDERIVATIVES:

$$\int e^x dx = e^x + C$$

$$\int a^x dx = a^x / \ln(a) + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int 1/(1+x^2) dx = \arctan x + C$$

$$\int 1/\sqrt{1-x^2} dx = \arcsin x + C$$

$$\text{CONSTANT MULT: } \int c \cdot f dx = c \cdot \int f dx$$

$$\text{SUM/DIFF: } \int (f \pm g) dx = \int f dx \pm \int g dx$$

## The Fundamental Theorem of Calculus [3]/[4]

$$\text{FTC PART 1: } d/dx \left[ \int_a^x f(t) dt \right] = f(x) \quad [3]$$

"derivative of an integral with variable upper limit equals the integrand"

$$\text{FTC PART 1 (chain rule version):} \quad [4]$$

$$d/dx \left[ \int_a^{u(x)} f(t) dt \right] = f(u(x)) \cdot u'(x)$$

$$\text{FTC PART 2: } \int_a^b f'(x) dx = f(b) - f(a) \quad [3]$$

"the definite integral of a derivative equals the NET CHANGE"

**KEY INSIGHT [4]** The Net Change interpretation is the core idea. If  $f'(t)$  is a rate (e.g., flow rate, velocity), then  $\int_a^b f'(t) dt$  is the total accumulation (e.g., total volume, displacement) over  $[a, b]$ . This is THE foundation of FRQ accumulation problems.

## U-Substitution — the chain rule in reverse [3]/[4]

The most important integration technique. Use when integrand is a product where one factor is the derivative of an inner function.

## RECIPE:

1. Identify  $u$  = inner function (often something inside another function)
2. Compute  $du = u'(x) dx$
3. Substitute to make integral entirely in terms of  $u$
4. Integrate w.r.t.  $u$
5. Substitute back  $x$

$$\text{Example: } \int 2x \cdot \cos(x^2) dx$$

$$\text{Let } u = x^2 \rightarrow du = 2x dx$$

$$\text{Integral becomes } \int \cos(u) du = \sin(u) + C = \sin(x^2) + C$$

## Definite Integrals — change limits when substituting [4]

For DEFINITE integrals with  $u$ -sub, change the limits to  $u$ -values. No back-substitution needed.

**Example.**  $\int_0^1 2x \cos(x^2) dx$ . Let  $u = x^2 \rightarrow du = 2x dx$ . When  $x=0$ ,  $u=0$ ; when  $x=1$ ,  $u=1$ . So  $\int_0^1 \cos(u) du = \sin(1) - \sin(0) = \sin(1)$ .

## Riemann Sums — definite integral as a limit [3]/[4]

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum f(x_i^*) \cdot \Delta x$$

where  $\Delta x = (b-a)/n$

## Common partitions:

- LEFT Riemann sum (LRAM): use  $f$  at LEFT endpoint of each subinterval
- RIGHT (RRAM): use  $f$  at RIGHT endpoint
- MIDPOINT (MRAM): use  $f$  at MIDPOINT
- TRAPEZOIDAL: average of L and R; better accuracy [4]

$$\text{When } f \text{ is INCREASING on } [a, b]: L < \int < R \quad (L \text{ underestimates})$$

$$\text{When } f \text{ is DECREASING on } [a, b]: R < \int < L \quad (R \text{ underestimates})$$

$$\text{When } f \text{ is CONCAVE UP on } [a, b]: \text{Trap} > \int > \text{Mid}$$

$$\text{When } f \text{ is CONCAVE DOWN: Mid} > \int > \text{Trap}$$

## UNIT 7 | DIFFERENTIAL EQUATIONS

## Slope Fields [3]/[4]

A slope field shows the slope  $\frac{dy}{dx}$  at each point in the plane. Drawing solution curves: start at a point and follow the slopes.

- [3] Sketch a few short line segments at each plotted point. Slope = value of  $\frac{dy}{dx}$  at that point.
- [4] Identify the differential equation a slope field came from. Test points: at (0,0), what's the slope? At (1,1)? Match to candidate equations.

## Separable Differential Equations [3]/[4]

SEPARABLE FORM:  $\frac{dy}{dx} = g(x) \cdot h(y)$

RECIPE:

1. Separate:  $\frac{dy}{h(y)} = g(x) dx$
2. Integrate:  $\int \frac{dy}{h(y)} = \int g(x) dx + C$
3. Solve for  $y$  (often involves taking exp of both sides)
4. Use INITIAL CONDITION to find  $C$

## Worked example — exponential growth [3]/[4]

**Problem.**  $\frac{dP}{dt} = kP$ ,  $P(0) = P_0$ . Find  $P(t)$ .

**Separate:**  $\frac{dP}{P} = k dt$ . **Integrate:**  $\ln|P| = kt + C_1$ . **Exponentiate:**  $P = e^{kt + C_1} = C \cdot e^{kt}$ . **Apply IC:**  $P(0) = C = P_0$ .

**Solution:**  $P(t) = P_0 \cdot e^{kt}$  — exponential growth ( $k > 0$ ) or decay ( $k < 0$ ).

## Logistic Growth (BC topic, but mentioned for AB context) [5]

$\frac{dP}{dt} = kP(M - P)$  [where  $M$  = carrying capacity]

Solution:  $P(t) = \frac{M}{1 + A \cdot e^{-kMt}}$  [A from initial condition]

Behavior:  $P$  grows fastest at  $P = M/2$  (the inflection point of growth curve)  
 $P$  approaches  $M$  as  $t \rightarrow \infty$

## Exponential Models — recognition rules [3]/[4]

- $\frac{dy}{dt} \propto y$  → exponential. Solution:  $y = y_0 e^{kt}$ .
- $\frac{dy}{dt} \propto (M - y)$  → bounded approach to  $M$ . Newton's law of cooling:  $\frac{dT}{dt} = -k(T - T_{\text{room}})$ .
- $\frac{dy}{dt} \propto y(M - y)$  → logistic. (BC topic)

**UNIT 8 | APPLICATIONS OF INTEGRATION · area, volume, average value****Area Between Curves [3]/[4]**

AREA between curves  $f$  and  $g$  on  $[a,b]$ , where  $f \geq g$ :  
 $A = \int_a^b [f(x) - g(x)] dx$  "top minus bottom"

If curves cross, split the integral at intersection points.

When integrating w.r.t.  $y$  instead of  $x$ :

$$A = \int_c^d [\text{right}(y) - \text{left}(y)] dy$$

**Volumes of Solids of Revolution [4]/[5]**

DISK METHOD (rotating about  $x$ -axis, no hole):

$$V = \pi \int_a^b [f(x)]^2 dx$$

WASHER METHOD (rotating about  $x$ -axis, with hole):

$$V = \pi \int_a^b ([\text{outer}(x)]^2 - [\text{inner}(x)]^2) dx$$

Rotation about  $y$ -axis or non-axis lines: shift functions and adapt.

KEY: visualize the cross-section. Disk = full circle. Washer = circle with hole.

**Volumes by Cross-Section [4]**

For solids whose cross-sections perpendicular to the  $x$ -axis are known shapes (squares, triangles, semicircles, etc.):

$$V = \int_a^b A(x) dx$$

where  $A(x)$  is the area of the cross-section at position  $x$ .

Examples (where the base of cross-section has width  $w(x)$ ):

Square cross-section:  $A(x) = w(x)^2$

Equilateral triangle:  $A(x) = (\sqrt{3}/4) \cdot w(x)^2$

Semicircle (diameter  $w$ ):  $A(x) = (\pi/8) \cdot w(x)^2$

**Average Value of a Function [3]/[4]**

AVG VALUE of  $f$  on  $[a,b]$ :

$$f_{\text{avg}} = (1 / (b-a)) \cdot \int_a^b f(x) dx$$

By Mean Value Theorem for Integrals: there exists  $c$  in  $[a,b]$   
 with  $f(c) = f_{\text{avg}}$ .

**Accumulation Functions — "so far" [4]**

$$G(x) = \int_a^x f(t) dt \quad \text{accumulation from } a \text{ to } x$$

Properties:

$$G'(x) = f(x) \quad (\text{FTC Part 1})$$

$G$  has critical points where  $f(x) = 0$

$G$  has inflection points where  $f'(x)$  changes sign (i.e.,  $f$  has extrema)

$G$  is increasing where  $f > 0$ , decreasing where  $f < 0$

**KEY INSIGHT [5] FRQ 1 is almost always an accumulation function problem with a graph of  $f(t)$  given.** Master the relationship between  $f$  and  $G$  — extrema of  $G$  occur where  $f$  crosses zero; inflections of  $G$  occur at extrema of  $f$ .



## PAGE 9 | FRQ STRATEGY · the 6 fixed FRQ types + universal discipline

### AP Calc AB FRQ Structure [3]

6 FRQs total: 2 calculator-allowed (Section IIA), 4 calculator-not-allowed (Section IIB). 90 min total. The 6 FRQ types are **ESSENTIALLY FIXED** across years:

| # | Type (Calculator?)                                 | What it tests                                                                            |
|---|----------------------------------------------------|------------------------------------------------------------------------------------------|
| 1 | Function defined by integral / Accumulation (calc) | Given graph of $f$ or $g(x) = \int_a^x f$ . Find values, derivatives, extrema, behavior. |
| 2 | Differential equation / Slope field (calc)         | Sketch slope field, solve separable ODE, use IC, interpret in context.                   |
| 3 | Functions defined by tables (no calc)              | Riemann sums, MVT applications, average rate of change vs derivative.                    |
| 4 | Particle motion (no calc)                          | Position, velocity, acceleration; speed up/slow down; distance vs displacement.          |
| 5 | Area + Volume (no calc)                            | Area between curves, volume by disks/washers, volumes by cross-section.                  |
| 6 | Implicit differentiation / Related rates (no calc) | $dy/dx$ implicitly; rate of change of one quantity in terms of another.                  |

### UNIVERSAL FRQ DISCIPLINE — non-negotiable [4]/[5]

- [3] **Show the SETUP** (the integral or formula) before computing. Even if calculator gives the number, the setup is what scores points.
- [3] **Use proper notation:**  $dy/dx$ ,  $f'(x)$ ,  $\int$ , limits of integration. NOT "the derivative" or "the integral."
- [3] **Carry units in context problems.** "32 ft/sec", "feet per second per second", "gallons per minute".
- [4] **Justify with theorems by NAME.** "By the IVT, since  $f$  is continuous on  $[1,3]$ ..." / "By the MVT, since  $f$  is continuous and differentiable..."
- [4] **Sign analysis tables** — when justifying max/min from  $f'$ , show a sign chart.
- [5] **Interpret in context** — after computing, give a sentence: "This means the volume is increasing at 32 gallons per minute at time  $t = 5$ ."
- [5] **Distinguish absolute vs local extrema.** State which one you're claiming.

### CALCULATOR USAGE — when allowed [4]

On calculator-allowed FRQs (1 and 2), use these 4 calculator features cold:

- Numerical derivative at a point — `nDeriv(f, x, value)`
- Numerical integral — `fnInt(f, x, a, b)`
- Solve equation / find zero — Solver or `zero()` feature
- Graph + intersect — visualize functions, find intersection points

Always show the setup first, then write "using calculator: [value]."

## PAGE 10 | THE 3/4/5 SCORE LADDER · tier-specific test-day

## THE SCORE LADDER — same problem, three tiers of answer

| 3-scorer writes...                       | 4-scorer writes...                                                                     | 5-scorer writes...                                                                                                                                                                                                                                                   |
|------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| " $f'(c) = 0$ so it's a critical point." | " $f'(c) = 0$ and $f'$ changes from + to – at $c$ , so by 1st DT, $c$ is a local max." | " $f'(c) = 0$ and $f'$ changes from + to – at $c$ (sign chart). By the <b>First Derivative Test</b> , $c$ is a local maximum. Since the only critical points on $[a,b]$ are $c$ and the endpoints, comparing values, $c$ is the <b>absolute maximum</b> by the EVT." |
| Computes $\int f(x) dx$                  | Shows setup + computes + writes "+ C" for indefinite                                   | Shows setup, computes, <b>interprets in context with units</b> : "This represents the total volume in gallons of water that flowed in over $[0, 4]$ "                                                                                                                |
| States the FTC                           | Applies FTC: $\int_a^b f'(x) dx = f(b) - f(a)$                                         | <b>Names the theorem explicitly</b> : "By the <b>Fundamental Theorem of Calculus Part 2</b> , ..." + interprets net change in context                                                                                                                                |
| Sets up related rates equation           | Sets up + differentiates implicitly w.r.t. $t$ + substitutes                           | Adds: <b>labeled diagram + units throughout + clear answer with sentence interpretation</b>                                                                                                                                                                          |
| Computes correct numerical answer        | Setup + answer + units                                                                 | Setup + computation + units + <b>sentence interpreting what the number means in context + mentions the relevant theorem when justifying</b>                                                                                                                          |

## TIER-SPECIFIC TEST-DAY CHECKLISTS

## For a 3 — bag ALL of these:

- ✓ MCQ: 22+ correct. Skip hard ones first; come back at the end. Never leave anything blank.
- ✓ Memorize derivatives + antiderivatives of standard functions (trig, exp, log).
- ✓ Power rule + product/quotient/chain rules cold.
- ✓ Each FRQ: ATTEMPT every part. Even partial setups earn points.
- ✓ FTC Part 2:  $\int_a^b f'(x) dx = f(b) - f(a)$ . Memorize.

## For a 4 — add these on top:

- ✓ MCQ: 28+ correct. Get comfortable with implicit differentiation + related rates.
- ✓ Each FRQ: setup → calc → answer. Show formulas symbolically before plugging in.
- ✓ Use first AND second derivative tests for extrema. State which one you're using.
- ✓ Sign analysis tables for  $f'(x)$  and  $f''(x)$ . Mark intervals.
- ✓ U-substitution: practice until automatic. Always show the substitution explicitly.
- ✓ Volumes: clearly identify if disks vs washers. Sketch the cross-section.

## For a 5 — add these on top:

- ✓ MCQ: 34+ correct.
- ✓ Every FRQ: cite theorems by name (IVT, EVT, MVT, FTC) when justifying.
- ✓ Distinguish local from absolute extrema. State which one and why.
- ✓ Interpret all answers IN CONTEXT with units in a complete sentence.
- ✓ Recognize accumulation functions:  $G(x) = \int_a^x f(t) dt$ . Master  $G$ 's behavior from  $f$ 's graph.
- ✓ Speed up vs slow down: SAME sign of  $v$  and  $a \rightarrow$  speed up. OPPOSITE  $\rightarrow$  slow down.
- ✓ Logarithmic differentiation when needed. L'Hôpital's rule for indeterminate limits.

**THE ONE THING THAT MATTERS MOST [3][4][5]: SHOW WORK + INTERPRET.** AP Calc grades the procedure as much as the final answer. After computing, ask: "Did I show the formula? Did I include units? Did I interpret what the number means in context?" Three yeses = full credit.

*If you do nothing else: pick your tier, study the matching badge tags, and apply the FRQ discipline checklist on test day. That's how you hit your target.*

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